



Daniel Ploug Hall

Portfolio: <https://daphall.github.io/>

Mobil: +45 60 15 26 04

E-mail: danielploug@hotmail.com

Profile

I am a Software Developer with 2 years of professional experience. I have applied my analytical approach to develop data pipelines, perform database development, create documentation and use my analytical approach to eliminate obscure bugs. In my programs, I prioritize usability and readability, as I am ultimately a customer of my code. I am a skilled verbal communicator and am strong with numerical mathematics. Feel free to visit my portfolio, to see some of my projects.

Professional qualifications

- **Languages:** Python, SQL, C (GCC), C++ (GCC)
- **Databases:** Postgres, Oracle
- **Python Libraries:** Psycpg3, OracleDB, Requests, Numpy, Scipy, Matplotlib, Seaborn, Pandas
- **Version Control:** Git, TortoiseSVN
- **Building Tools:** Make, CMake
- **Documentation Tools:** Sphinx
- **Data science:** Curve fitting, Data visualization, Data analysis
- **Parallel interfaces:** OpenMP, MPI
- **Numerical methods:** Finite volume method, Finite difference method, Numerical optimization, Leapfrog integration, Gaussian Quadrature

Personal qualifications

- Verbal Communicator
- Independent worker
- Helpful
- Detail orientated
- Curious

Working Experience

Software developer at Septima - Nov 2023 - Aug 2024

- Development of APIs, used by their web service
- Development of the main database such as implementing new features.
- Ad hoc assignments for costumers, which include but not limited to route calulations

PhD student at Dansk brand- og sikrings institut and KU - Jun 2022 - May 2023

- Research into moisture transport in fire retarded wood
- Balancing KU and DBIs interest in the project

Postgres and Python developer at Danmark Statistik - Nov 2021 - Jun 2022

- Custom datapipelines from files into Postgres
- Developing Postgres and Oracle databases
- Development of an easy to use privileges system in postgres
- Solved a years long problem with PyInstaller and MKL

TA at Datalogisk Institut (DIKU, KU) in Numerical Methods - Sep - Nov 2020

- Teaching 20+ student

TA at Datalogisk Institut (DIKU, KU) i Computational Methods in Simulation - Apr - Jun 2021

- Grading assignments

TA at the Niels Bohr Institute in High-Performance Parallel Computing

- Feb - Apr 2021

- Teaching 15+ people

Education - University of Copenhagen (KU)

Master in Physics with a specialty in Computational Physics - Sep 2019 - Jun 2021

- Average grade: 10.2
- Thesis in "Dust dynamics in a collapsing Prestellar core" under Troels Haugbølle
- Developed a post-processing dust simulation for one of my supervisors codebases

Bachelor In Physics with a speciality in Astrophysics - Sep 2016 - Jun 2019

- Average grade: 7.4
- Bachelor Project in "Dust dynamics in a protoplanetary disk" under Troels Haugbølle
- Recreated a 1D protoplanetary disk simulation with dust coagulation

About me

In my spare time, you can find me at my club archery range, practising my archery skills and making medieval jokes with a friend of mine. Furthermore, I use a lot of my spare time on tabletop role-playing games, where I create stories with my friends and engage in game design. This passion led me to develop a dice modelling library for Python called "dicechanics", which I use on a semi-daily basis. I also love nature and once a year go hiking with my friends.